INCH-POUND

MIL-T-27/187B
13 March 1991
SUPERSEDING
MIL-T-27/187A
21 AUGUST 1986

MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN, AUTOTRANSFORMER, TF5TO3ZZ

Ⓑ Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.

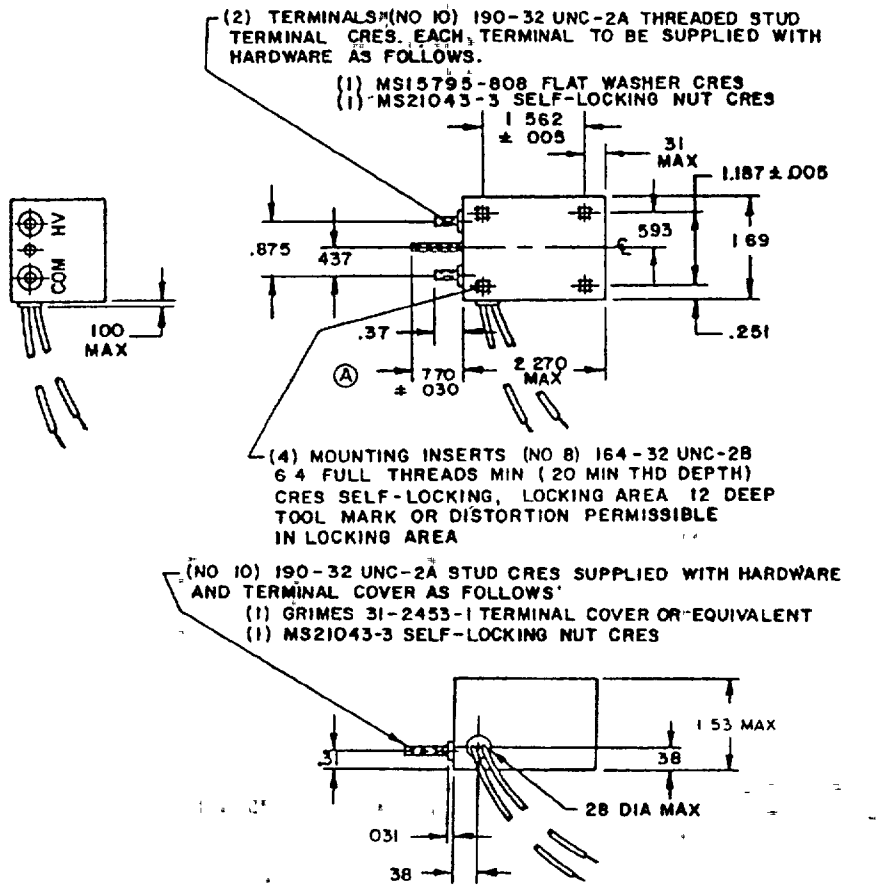
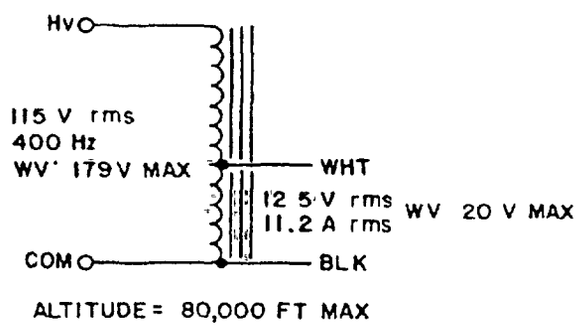


FIGURE 1 Dimensions and configurations.

Ⓑ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:47Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm
.005	0.13	.251	6.38	.770	19.56
.031	0.79	.28	7.1	.875	22.22
100	2.54	.31	7.9	1.187	30.15
.164	4.17	.37	9.4	1.53	38.9
.190	4.83	.38	9.7	1.562	41 96
.20	5.1	.437	11.10	1.69	42.9
		.593	15.06	2.270	57.69

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place dimensions and ±.03 (0.8 mm) for two place dimensions.
4. Marking shall be on the top of the case.
5. Stud shoulder shall form an integral part of the threaded stud terminal.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:47Z
Check the source to verify that this is the current version before use.

REQUIREMENTS (When designations or colors in parentheses, i.e., (COM-HV) (BLK-WHT) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings

Primary (COM-HV) 173 volts ampere, 115 volts rms, 400 hertz

Secondary (BLK-WHT) 12.5 volts rms, 11.2 amperes rms

Working voltage.

(COM-HV): 179 volts maximum.
(BLK-WHT): 20 volts maximum.

DC resistance: At ambient temperature of +20°C

(BLK-WHT). 0.057 ohm ±15 percent.
(WHT-HV). 1.45 ohms ±15 percent.

Design and construction.

Dimensions and configuration: See figure 1.

Duty cycle. Continuous.

Case: Diallyl phthalate (MIL-P-19833, type GDI-30F).

Weight: 0.820 pound maximum.

Terminal type:

Primary. Corrosion-resistant steel, threaded integral shoulder-stud.

Secondary. Number 18 AWG insulated lead wire in accordance with MIL-W-16878, type EE.

Terminal height.

Primary: See figure 1.

Terminal length:

Secondary. 16 ±1 inch.

Altitude 80,000 feet maximum.

Operating temperature range: -54°C to +170°C.

Operating temperature: +170°C maximum.

Insulating materials.

Magnet wire insulation. Modified polyester base, polyamide-imide overcoat, +200°C.

Encapsulating material Epoxy, +170°C

Paper Aramid, +220°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:47Z
Check the source to verify that this is the current version before use.

Tape:

Backings Polyamide film, glass cloth, aramid paper, +220°C.

Adhesive Silicone, +180°C.

Bobbin Aramid, +220°C.

Housing. Diallyl phthalate, +180°C.

Terminal strength. MIL-STD-202, method 211, test condition A, 5 pounds.

Vibration, high frequency: MIL-STD-202, method 204, test condition D.

Salt spray (corrosion): MIL-STD-202, method 101, test condition B.

Dielectric withstanding voltage

At sea level: 500 volts rms

At barometric pressure: 300 volts rms.

Electrical characteristics

No load. With 115 volts rms, 400 hertz across (COM-HV).

Current in (COM-HV): 150 milliamperes maximum.

Power in (COM-HV): 2.9 watts maximum.

Voltage across (BLK-WHT): 13.7 volts rms ±2 percent.

Rated load With 115 volts rms, 400 hertz across (COM-HV).

Voltage across (BLK-WHT). 12.5 -0, +0.4 volts rms at an ambient temperature of +70°C.

Voltage across (BLK-WHT) 12.5 -0, +0.7 volts rms at an ambient temperature of +25°C.

Temperature rise: +100°C with 115 volts rms, 400 hertz across (COM-HV) at an ambient temperature of +70°C.

Voltage regulation: 10 percent maximum.

Ⓑ Quality assurance provisions.

Qualification inspection Not applicable

Quality conformance inspection Groups A and B tests of MIL-T-27 shall be applicable.

Marking location: See figure 1.

Part number: M27/187-01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:47Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities:
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent:
DLA - ES

(Project 5950-0753-40)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:47Z
Check the source to verify that this is the current version before use.